

Miguel Montalvo Villarreal

A00839068

LINK COLLAB

https://colab.research.google.com/drive/18dUdmAfgQ2pnS5IRQrwTWVGqUeJr1p_T?usp=sharing

LINK PROMPT:

https://grok.com/share/c2hhcmQtMg%3D%3D_290ede63-fdf8-4d75-8347-dc0acb525e85

LINK GITHUB:

<https://github.com/miguelmv05/EXAMEN-SERIES-DE-TIEMPO.git>

INTERPRETACIONES:

1. Stationarity Analysis

- **ADF Test:**
 - **Statistic:** -0.4674, **p-value:** 0.8982
 - **Interpretation:** The p-value (0.8982) is much greater than 0.05, so we fail to reject the null hypothesis of a unit root. This suggests that the Money Supply series is likely **non-stationary** at its level form. The ADF statistic is also higher (less negative) than the critical values at 1%, 5%, and 10%, reinforcing the non-stationarity conclusion.
- **KPSS Test:**
 - **Statistic:** 3.6046, **p-value:** 0.0100
 - **Interpretation:** The p-value (0.0100) is less than 0.05, so we reject the null hypothesis of stationarity. The KPSS statistic exceeds all critical values (1%, 5%, 2.5%, and 10%), confirming that the Money Supply series is **non-stationary**.
- **Summary:** Both the ADF and KPSS tests consistently indicate that the Money Supply series is **non-stationary** in its level form. This implies that the series likely exhibits trends or other non-stationary behaviors, requiring differencing to achieve stationarity for modeling purposes.

2. Correlogram Analysis

- The correlogram for the first 10 lags shows the **Autocorrelation Function (ACF)** and **Partial Autocorrelation Function (PACF)**:

- **ACF:** The ACF values are very high (close to 1) and decay slowly from 0.996 at lag 1 to 0.955 at lag 10. This slow decay is a hallmark of a non-stationary series, as stationary series typically show faster decay in ACF.
- **PACF:** The PACF shows a significant spike at lag 1 (0.998) and then drops to small, mostly negative values close to zero for higher lags. This suggests that an autoregressive process of order 1 or 2 might be appropriate, as the PACF cuts off after the first lag.
- **Interpretation:** The slow decay in ACF and the significant spike in PACF at lag 1 confirm the non-stationarity of the series and suggest that differencing (likely one difference) is needed to make the series stationary. The PACF behavior also supports including autoregressive terms in the ARIMA model.

3. ARIMA Model

- **Best Model:** ARIMA(2,1,3)
 - **p=2:** Two autoregressive terms, indicating that the current value of the differenced series depends on the previous two values.
 - **d=1:** One difference is required to make the series stationary, consistent with the non-stationarity findings from the ADF and KPSS tests.
 - **q=3:** Three moving average terms, suggesting that the model accounts for three lagged forecast errors.
 - **AIC:** 2298.30, which indicates the model's goodness of fit (lower AIC values suggest better models, though comparisons with other models would be needed for context).
- **Interpretation:** The ARIMA(2,1,3) model is appropriate given the non-stationary nature of the Money Supply series (requiring $d=1$) and the ACF/PACF patterns. The inclusion of two AR terms aligns with the significant PACF spike at lag 1 and potential influence at lag 2, while the three MA terms help capture additional dynamics in the residuals.

4. Forecast Interpretation

- **Last Observed Value:** 1093.46
- **Average Forecast Value:** 1065.22 (over the forecast horizon)
- **Forecast Change:** -28.24 (average decrease from the last observed value)
- **Trend:** Downward forecast trend
- **95% Confidence Interval at Period 30:** [908.60, 1162.12]
- **Next 5 Periods Forecast:**
 - 489: 1089.50
 - 490: 1089.28
 - 491: 1088.75
 - 492: 1087.93
 - 493: 1086.85
- **Interpretation:**
 - The forecast predicts a **downward trend** in Money Supply over the next periods, with values gradually decreasing from 1089.50 to 1086.85 over the next five periods. This suggests a slight contraction or slowdown in the growth of Money Supply.

- The average forecast value (1065.22) is lower than the last observed value (1093.46), indicating an expected decline of approximately 28.24 units on average.
- The **95% confidence interval** at period 30 ([908.60, 1162.12]) is relatively wide, reflecting uncertainty in the long-term forecast. This suggests that while the point forecast indicates a downward trend, there is significant variability, and the actual Money Supply could deviate substantially from the predicted value.
- The small, consistent declines in the five-period forecast suggest a stable but gradual downward movement, which may reflect economic factors such as tighter monetary policy, reduced liquidity, or other macroeconomic conditions affecting Money Supply.

5. Economic and Practical Implications

- **Non-Stationarity:** The non-stationary nature of the Money Supply series indicates that it has a trend or other persistent patterns. This is typical for macroeconomic variables like Money Supply, which often grow over time due to inflation, economic growth, or policy changes. The need for one difference ($d=1$) in the ARIMA model confirms that the series can be made stationary by differencing, which is common in time series modeling of monetary aggregates.
- **Downward Forecast Trend:** The forecasted decline in Money Supply could have significant economic implications:
 - A reduction in Money Supply may signal tighter monetary policy, potentially aimed at controlling inflation or stabilizing financial markets.
 - It could also reflect reduced economic activity, lower - A declining Money Supply might lead to higher interest rates, reduced consumer spending, or slower economic growth in the short term.
 - Policymakers and central banks may need to monitor this trend closely, as prolonged declines in Money Supply could exacerbate deflationary pressures or economic slowdowns.
- **Model and Forecast Reliability:** The ARIMA(2,1,3) model appears well-suited to the data, but the wide confidence interval suggests caution in relying heavily on long-term forecasts. External factors (e.g., monetary policy changes, fiscal stimulus, or global economic conditions) could alter the trajectory of Money Supply, which the model may not fully capture.

6. Recommendations

- **Further Analysis:** Consider testing for structural breaks or seasonality in the Money Supply series, as these could affect stationarity and model fit. Additionally, explore alternative models (e.g., SARIMA, VAR, or machine learning approaches) to compare forecasting performance.
- **Policy Monitoring:** Given the downward forecast, central banks or policymakers should assess whether this aligns with their monetary policy objectives (e.g., inflation targeting) and adjust policies if necessary.

- **Incorporate Exogenous Variables:** If available, include exogenous variables (e.g., interest rates, GDP growth, or inflation) in the model to improve forecast accuracy and capture macroeconomic dynamics.
- **Short-Term Focus:** Given the uncertainty in the long-term forecast (wide CI), focus on the short-term (next 5–10 periods) predictions for decision-making, as these are likely more reliable.

Conclusion

The Money Supply series is non-stationary, as confirmed by both ADF and KPSS tests, and requires one difference to achieve stationarity. The ARIMA(2,1,3) model effectively captures the series' dynamics, forecasting a downward trend with an average decline of 28.24 units from the last observed value. This suggests a potential contraction in Money Supply, which could have broader economic implications. However, the wide confidence interval indicates uncertainty, particularly for longer horizons. Policymakers and analysts should use these forecasts cautiously, monitor external economic conditions, and consider additional modeling approaches to refine predictions.

INTERPRETACIONES PERSONALES

El análisis muestra que el Money Supply no es estacionario, lo que significa que su comportamiento cambia con el tiempo y no sigue un patrón constante. En términos económicos, una serie estacionaria sería aquella en la que la cantidad de dinero en circulación se mantiene relativamente estable, con fluctuaciones menores que no afectan la tendencia general. Sin embargo, al observar los resultados de las pruebas ADF y KPSS, podemos concluir que el Money Supply no cumple con esta estabilidad, lo que sugiere que factores externos influyen en su movimiento, ya sean políticas monetarias, ciclos económicos o eventos imprevistos.

Un Money Supply no estacionario puede implicar que la economía está atravesando períodos de expansión o contracción que afectan la cantidad de dinero disponible. La falta de estacionariedad también significa que, para realizar análisis y pronósticos precisos, es necesario aplicar transformaciones como la diferenciación para estabilizar la serie y hacerla más predecible.

El correlograma muestra que los valores actuales del Money Supply tienen una alta correlación con sus valores pasados. Esto indica que los cambios en la cantidad de dinero en circulación no ocurren de manera abrupta ni aleatoria, sino que están influenciados por su comportamiento anterior. En términos simples, si el Money Supply ha sido alto en los últimos meses, es probable que siga siendo alto por un tiempo antes de experimentar cambios significativos.

Este fenómeno es importante porque nos dice que el Money Supply no está sujeto a variaciones drásticas sin una causa específica. En otras palabras, no es común que la cantidad de dinero en la economía cambie repentinamente sin un factor estructural detrás, como ajustes en las tasas de interés, inflación o decisiones de política monetaria por parte del banco central.

Uno de los hallazgos clave del modelo ARIMA es que el Money Supply presenta una leve tendencia a la baja. El último valor registrado en la serie fue 1093.46, pero el pronóstico promedio indica que podría reducirse a 1065.22 en los próximos períodos. Aunque esta disminución no es abrupta, sí señala una tendencia que podría tener implicaciones económicas.

Una reducción en el Money Supply suele estar relacionada con una política monetaria restrictiva, donde las autoridades buscan controlar la inflación o reducir el crecimiento del crédito. También puede reflejar una disminución en la demanda de dinero, lo que podría estar vinculado a una desaceleración en el consumo y la inversión. Si esta tendencia persiste, podría afectar la liquidez del mercado y las decisiones económicas de empresas y consumidores.

Por otro lado, es importante destacar que los pronósticos tienen cierto grado de incertidumbre. El intervalo de confianza al período 30 muestra que el Money Supply podría situarse en un rango entre 908.60 y 1162.12, lo que indica que la tendencia no es completamente predecible y está sujeta a cambios dependiendo de las condiciones económicas.

El AIC del modelo es 2298.30, lo que indica que es una opción estadísticamente adecuada en comparación con otros modelos alternativos. Sin embargo, como cualquier modelo de predicción, los resultados deben interpretarse con precaución. El intervalo de confianza nos da una idea de la variabilidad de las estimaciones y nos recuerda que las condiciones económicas pueden cambiar y afectar los pronósticos.

La reducción del Money Supply puede tener efectos en varias áreas de la economía:

Política monetaria: Si el banco central está reduciendo la cantidad de dinero en circulación, es posible que busque controlar la inflación o ajustar la actividad económica.

Mercados financieros: Menos dinero en circulación puede significar menor liquidez, lo que podría afectar la disponibilidad de crédito y las tasas de interés.

Consumo y producción: Si las empresas y consumidores tienen acceso a menos dinero, podrían reducir su actividad económica, afectando el crecimiento.

Si la tendencia descendente se mantiene, podría ser señal de una política monetaria restrictiva o de una menor demanda de dinero en la economía. Es importante monitorear los próximos datos para evaluar si esta reducción se profundiza o si el Money Supply vuelve a estabilizarse.